

PROJECT PROPOSAL

BLOCK 1D-2024-2025

Market Event Predictor with News Sentiment Analysis

Group 13

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I. Introduction

The "Market Event Predictor with News Sentiment Analysis" project aims to develop an intelligent system designed to forecast short-term stock market anomalies, including events like crashes, dips, and rallies, by analysing minute-by-minute NASDAQ-100 price data and incorporating financial news sentiment. The primary objective is to equip Move Tickers with an advanced predictive tool that supports their strategic expansion into wealth management and their exploration of high-frequency trading, enabling the timely identification of unexpected market movements and the generation of actionable alerts. In the broader financial sector, leveraging machine learning for predictive analytics from complex datasets is becoming essential for maintaining a competitive edge. Consequently, the application of machine learning is critical for this project as it offers the capability to discern subtle and complex patterns in market data that traditional analysis might overlook, thereby providing Move Tickers with enhanced risk management capabilities and the potential to capitalize on emergent trading opportunities, directly aligning with their goal to integrate sophisticated quantitative models for improved market performance and client service.

II. Problem Statement

Move Tickers is strategically expanding into the wealth management sector and exploring advanced trading methodologies like low-latency and high-frequency trading (HFT). A core challenge in this endeavor is the timely identification and prediction of "out-of-pattern, or unexpected price or volume movements". Current mainstream financial dashboards often provide only basic information, such as 52-week highs and lows, failing to offer the predictive insights necessary for proactive decision-making in dynamic markets. The stock market's behavior is influenced by a multitude of complex, interconnected factors, making it exceedingly difficult to anticipate significant events like sudden crashes, dips, or rallies. This leaves traders and asset managers vulnerable to being caught off guard by adverse movements or missing out on fleeting opportunities.

Solving this problem is crucial for Move Tickers as it directly supports their ambition to integrate advanced quantitative models and leverage machine learning to uncover hidden patterns in market data. An AI-driven solution capable of predicting market anomalies offers significant business value, including:

- **Enhanced Decision-Making:** Providing traders and fund managers with timely alerts on predicted market events allows for faster, more informed trading decisions.
- **Improved Risk Management:** Early warnings about potential market downturns, such as crashes or dips, can help mitigate substantial financial losses.
- **Competitive Advantage:** Uncovering subtle market patterns that elude traditional analysis can lead to superior trading strategy performance and a distinct edge in the market.

- **New Service Offerings:** The developed system can be integrated into existing HFT solutions to boost ROI, offered as a standalone subscription service, or provided as a white-label solution for other platforms.

Project Goals and Objectives:

The primary goal of this project is to develop a "Market Event Predictor with News Sentiment Analysis" system. This system will aim to forecast short-term stock market anomalies for NASDAQ-100 securities by combining price data analysis with financial news sentiment.

Key Tasks and Expected Outcomes:

- **Data Collection and Preprocessing:** Gather and prepare minute-by-minute NASDAQ-100 stock price data and financial news data from external aggregators. This includes cleaning, feature engineering, and aligning time-series price data with textual news data.
- **Sentiment Analysis:** Implement NLP techniques to extract and quantify sentiment from financial news articles, linking it to market movements.
- **Machine Learning Model Development:** Design, train, and evaluate machine learning models (e.g., classification or anomaly detection models) to predict defined market events such as crashes, dips, and rallies.
- **Prediction Output:** The system will output predictions of market events within a specified time horizon, potentially including confidence scores and other relevant statistics.
- **System Implementation:** Develop a prototype, which could be a standalone dashboard or an API, to deliver these predictions and insights to end-users.
- **Evaluation:** Rigorously evaluate the model's performance using metrics like F1-score, precision, recall (especially for rare events), false positive rate, and prediction latency. Backtesting against historical data will also be crucial.

These objectives align directly with Move Tickers' broader business needs by providing a concrete tool to address their question of "When should we buy the dip?". It also supports their goal of developing predictive warning systems for unexpected market events and leveraging ML to uncover patterns that are not immediately apparent through conventional analysis. The project will provide actionable insights, contributing to Move Tickers' data-driven approach to financial markets.

III. Data Description

For the "Market Event Predictor with News Sentiment Analysis" project, we plan to utilize two primary types of data:

1. High-Frequency Stock Market Data:

- **Source:** Historical minute-by-minute trading data for NASDAQ-100 constituent stocks (using AAPL as an initial representative example).
- **Structure & Key Attributes (Initial):** Time-series data indexed by datetime (YYYY-MM-DD HH:MM:SS), containing Open, High, Low, Close (OHLC) prices, Volume, and Transaction counts.
- **Planned Engineered Features (from Market Data):** An extensive set of features will be engineered. This includes:
 - Lagged price/volume, returns over various periods.
 - Moving Averages (SMA, EMA) and price-to-MA ratios.
 - Volatility measures (ATR, Bollinger Bands, Realized Volatility, Volatility of Volatility).
 - Momentum indicators (RSI, MACD, ROC).
 - Trend indicators (Ichimoku Cloud, MA slopes).
 - Volume-based features (VWAP, intraday volume z-scores).
 - Time-based features (cyclical encodings of hour/day/month, market open/close flags).
 - Interaction features (e.g., Dollar Volume).
 - Long-term context features (52-week high/low, 30D Avg Volume, etc.) will be derived from supplementary daily stock data to ensure robustness and avoid missing values associated with long rolling windows on minute data.
- **Target Variable (event):** A custom multi-class column (e.g., 'normal', 'dip', 'flash_crash') will be engineered based on rule-based detection of future price action relative to each minute.

2. Financial News Data (optional):

- **Source:** External news aggregators and APIs, focusing on NASDAQ-100 relevant news.
- **Structure & Key Attributes (Expected):** Textual data including publication timestamp, headline, and article content.
- **Planned Engineered Features (from News Data):** Natural Language Processing (NLP) will be applied to extract:
 - Sentiment scores (positive/negative/neutral).
 - Subjectivity scores.
 - Possibly topic models or named entities relevant to market movements.
 - These features will be aggregated over time windows and aligned with the minute-level market data.

Potential Data Quality Issues and Handling Strategy (Planned):

• Market Data:

- *Missing Values:* Raw data will be checked for completeness. NaNs introduced during feature engineering (e.g., rolling window warm-ups) will be handled systematically, likely via forward/backward filling or by ensuring sufficient historical data for calculations. The use of daily data

for long-term features aims to minimize initial NaNs for those specific columns.

- *Outliers*: EDA will be performed to identify and investigate potential data errors or extreme outliers in price/volume.
- *Timestamp Consistency*: Data will be parsed with datetime indexing and sorted chronologically.
- **News Data:**
 - *Noise/Relevance*: Filters will be developed to select financially relevant news.
 - *Timestamp Alignment*: Critical attention will be paid to accurately align news publication times with market data, considering potential lags.
 - *Sentiment Analysis Challenges*: Robust NLP models, potentially fine-tuned on financial text, will be explored to handle the nuances of financial language.
 - *Duplicates/Volume*: Strategies for deduplication and efficient processing of high-volume news streams will be implemented.

This combined dataset, rich with both market-derived technical features and news-derived sentiment features, will form the input for our predictive models. The primary target (event) will be engineered to reflect distinct, actionable market phenomena.

IV. Methodology

Our team will adopt a structured machine learning approach to develop the "Market Event Predictor with News Sentiment Analysis" system. The process will encompass data collection and preprocessing, model selection and training, rigorous evaluation, and a clear deployment strategy.

Data Collection & Pre-processing

This foundational phase focuses on acquiring, cleaning, transforming, and integrating the diverse datasets required to train and validate our predictive models. The primary data streams are historical stock market data and financial news data.

1. Stock Market Data Collection:

- **Minute-Frequency Data**: The core market dataset will consist of historical 1-minute frequency trading data for NASDAQ-100 constituent securities. Initially, Apple Inc. (AAPL) data (from AAPL_minute_data.csv, spanning approximately 10 years) will be used for development and a full feature engineering pipeline. This data includes Timestamp, Open, High, Low, Close (OHLC) prices, Volume, and Transaction counts.
- **Daily-Frequency Data**: To incorporate long-term market context robustly, daily historical OHLCV data for the corresponding securities (e.g., AAPL.csv) will also be collected.

2. Financial News Data Collection: (optional)

- Financial news articles, headlines, and associated metadata (publication timestamp, source) relevant to NASDAQ-100 companies will be gathered from selected external news aggregators and financial news APIs (e.g., NewsAPI, Alpha Vantage, or similar publicly available/licensed sources). The selection will prioritize sources with comprehensive coverage, historical depth, and reliable timestamping.

3. Market Data Pre-processing & Feature Engineering:

- **Loading and Initial Cleaning:** Both minute and daily stock data will be loaded, parsed with Datetime indexing, sorted chronologically, and inspected for any obvious errors or inconsistencies.
- **Long-Term Feature Calculation (using Daily Data):** Key long-term indicators such as 52-week highs/lows, 30-day average daily volume, 100-day moving averages, and 30-day price Z-scores will be calculated from the daily stock data. These daily metrics will then be mapped (forward-filled) to the corresponding minute-level data to provide each minute with its relevant daily context. This approach mitigates issues of NaN values for long-window features if calculated directly on potentially sparse minute data over extended calendar periods.
- **Minute-Level Feature Engineering:** An extensive set of technical features will be engineered from the minute-level data. This includes:
 - *Basic Transformations:* Lagged price and volume data, minute-by-minute returns, percentage changes.
 - *Moving Averages:* Simple Moving Averages (SMA) and Exponential Moving Averages (EMA) over various short to medium-term windows (e.g., 50-min, 200-min), and ratios of price to these MAs.
 - *Volatility Indicators:* Average True Range (ATR), Bollinger Bands (including width and %B), realized volatility over different short-term windows (e.g., 5-min, 30-min), and Volatility of Volatility (VoV).
 - *Momentum Indicators:* Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD, Signal, Histogram), Rate of Change (ROC).
 - *Trend & Other Price-Derived:* Ichimoku Cloud components, MA slopes, intraday amplitude, overnight gaps, Volume-Weighted Average Price (VWAP).
 - *Volume-Based:* Intraday volume Z-scores (comparing to typical volume for that specific time of day), volume acceleration, high volume bar flags.
- **Advanced Time-Based Features:** Cyclical encoding (using sine/cosine transformations) of time components (hour, day of week, month of year), flags for market open/close periods, and minutes remaining until market close.
- **Interaction Features:** Creation of new features by combining existing ones (e.g., Dollar Volume = Close * Volume, RSI * Volume_pct_change).
- **Handling Missing Values:** NaNs generated during feature calculation (e.g., initial warm-up for rolling windows) will be handled primarily

through forward-fill (ffill) followed by back-fill (bfill) to ensure a complete feature set.

4. **News Data Pre-processing & Sentiment Analysis:**

- **Text Cleaning:** Raw news text (headlines and articles) will be cleaned by removing HTML tags, special characters, boilerplate content, and other noise.
- **Deduplication:** Duplicate or near-duplicate news items from different sources will be identified and removed.
- **Sentiment Scoring:** Natural Language Processing (NLP) techniques will be applied to quantify the sentiment of each news item. This may involve lexicon-based approaches (e.g., VADER) or machine learning models (e.g., FinBERT, if feasible) fine-tuned on financial text to classify sentiment as positive, negative, or neutral, and potentially assign a continuous sentiment score. Subjectivity analysis may also be performed.
- **Feature Creation:** The raw sentiment scores will be transformed into features. This will likely involve aggregating sentiment over specific time windows (e.g., average sentiment of news related to a stock in the 15, 30, or 60 minutes leading up to a market data point) and potentially creating features based on sentiment shifts or volume of news.

5. **Data Integration and Target Variable Creation:**

- **Timestamp Alignment:** Engineered news sentiment features will be carefully aligned with the minute-level market data based on timestamps. This is a critical step to ensure that news information precedes the market data point it's associated with for predictive purposes.
- **Target Variable (event column):** The multi-class target variable will be generated on the minute-level market data. As detailed in project planning, this involves a rule-based system (detect_events_enhanced function) using an event_categories dictionary to label each minute as 'normal', 'flash_crash', 'crash', 'volatility_spike', 'rebound_rally', or 'dip', based on defined price/volume movements and recovery criteria in subsequent time windows.
- **Final Dataset Assembly:** The processed market features and news sentiment features will be combined into a single, comprehensive dataset (AAPL_consolidated_features_events_final.csv) ready for model training.

This meticulous data collection and pre-processing pipeline is designed to create a high-quality, feature-rich dataset that maximizes the potential for the subsequent machine learning models to identify predictive patterns for market anomalies.

Model Selection & Training

- **Model Candidates:** We plan to explore several machine learning models suitable for time-series classification and anomaly detection:
 - **Supervised Classification Models:** Random Forest, XGBoost, and deep learning models like Long Short-Term Memory Networks (LSTMs), Transformers, or Temporal Convolutional Networks (TCNs). These models are chosen for their ability to capture complex, non-linear patterns, and temporal dependencies in the data.
 - **Anomaly Detection Models:** These may be employed for identifying rare and sudden events like flash crashes.

The choice between supervised/semi-supervised and unsupervised approaches will depend on the effectiveness of the labeling strategy and initial model performance. We may train one model per company or a general model with the ticker ID as an input feature.

- **Justification:** These models are well-suited for handling the sequential nature of financial time series and integrating numerical price data with textual sentiment features. LSTMs and Transformers, for example, are adept at learning long-range dependencies, which can be crucial for market prediction.
- **Data Splitting:** The data will be split chronologically into training, validation, and testing sets. This is critical for time-series data to prevent data leakage (i.e., using future information to train the model) and to simulate real-world prediction scenarios.

Evaluation Metrics

To assess model performance rigorously, we will use a combination of standard machine-learning metrics and business-relevant indicators:

- **Classification Metrics:**
 - **F1-score:** A harmonic mean of precision and recall, providing a balanced measure, especially for imbalanced classes.
 - **Precision:** The proportion of correctly predicted positive events out of all predicted positive events.
 - **Recall (Sensitivity):** The proportion of actual positive events that were correctly identified. This is particularly important for rare but critical events like market crashes.
 - **False Positive Rate:** The proportion of negative events incorrectly classified as positive. Minimizing this is crucial to avoid triggering false alarms and eroding user trust.
- **Backtesting:**
 - The model's accuracy will be evaluated through backtesting against historical data to assess its ability to catch known historical market events like crashes and dips. This will also involve metrics like Return, Sharpe Ratio, and Win Rate if simulating trading strategies.

Deployment

The final model and its outputs will be made accessible to Move Tickers through one or more of the following channels:

- **Standalone Dashboard/Application:** A user-friendly dashboard (potentially using Gradio for an MVP) could be developed to display predictions, market event alerts, and other generated statistics like prediction confidence or stock insights. This would provide a direct interface for traders and analysts.
- **REST API:** A REST API could be created to allow for easier integration of the prediction system with Move Tickers' existing internal tools, and client platforms, or potentially as a white-label solution for other financial services.

The deployment will be phased, starting with a prototype and iteratively improving based on feedback and performance. The system architecture will consider scalability and reliability to handle real-time data ingestion and prediction generation.

V. Project Timeline

- **This week (effectively by Wednesday, May 28th, 2025):** Complete the Exploratory Data Analysis (EDA).
- **Next Tuesday (June 3rd, 2025):** Have the final data table ready for review.
- **By Week 5:** Complete model building.
- **Week 6:** Focus on backtesting and creating a UI or a wrapper for the project.
- **Week 7:** Collate all work and submit the project.

VI. Reference

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Disclaimer: ChatGPT was used solely to edit and refine text written by the authors.